



Fantastic Beasts 2's villain

The sequel to the Harry Potter spinoff movie has casted Iron Fist actor David Sakurai as the villain. <http://dgit.in/FntstcBsts2V>



PUBG unhappy with Fortnite

After Fortnite announced a standalone game following the battle royale genre, PUBG isn't really happy. <http://dgit.in/PUBGFrnt>

shown by major scientific establishments. CERN, one of the most well-established nuclear physics institutions in the world, held a colloquium on LENR in 2012 that discussed theoretical and experimental LENR progress since 1989. ARPA-E, the DARPA equivalent for energy research, added LENR to the list of fields that will be funded by their funding program. MIT, who were one of the major critics of the 1989 experiment, are now offering a course on cold fusion. NASA and Boeing have teamed up to look at the possibility of LENR to power future green-powered aircrafts. Countries like India, Japan, Italy and the US have set up facilities to conduct research on cold fusion. Apart from these developments in research, there have been attempts at commercialising LENR in recent years. The most infamous attempt being, Andrea Rossi and his E-Cat.

SCHRODINGER'S E - CAT

E-Cat (Energy Catalyser) is an alleged cold fusion reactor invented by Italian inventor, Andrea Rossi. In 2011, the first public demonstration of the reactor was held in an industrial plant in Bologna. The reactor, compact enough to be placed on a table, released 2,635 KWH of energy in a total of five hours. To put things into perspective, the annual power consumption of an average Indian household is only around 900 KWH per year. Too good to be true? That's what many of the leading scientists believed. The members present at the demonstration were not allowed to check the measurements themselves, which led to suspicion that Rossi manipulated values. Andrea Rossi, known to be eccentric, has been extremely secretive of his invention, neither revealing the physics behind the reactor nor allowing scientists to examine his product thoroughly. He communicates with the outside world, only through his blog, The Journal Of Nuclear Physics.

A second demonstration of the E-Cat was conducted in 2014, in the presence of researchers from The University of Upsala and The University of Bologna.

The E-Cat was run for an extended period to prove that the energy could not have been created by external sources. Their report (a.k.a The Lugano report) states, "The total net energy obtained during the 32 days run was about 1.5 MWh. This amount of energy is far more than can be obtained from any known chemical sources in the small reactor volume." The number of followers for Rossi increased, but a few refused to believe, questioning the extent of freedom of the researchers.

The general consensus amongst the scientific community about the validity of his claims is mostly negative. This is a reasonable verdict because, firstly, he has been very secretive about the work-



Andrea Rossi with the E-Cat

ings of his product and not very liberal with his demonstrations. Secondly, Rossi has been convicted of white-collar criminal charges for his earlier business adventures. He has also had a contract with the US army for making heat generating devices, but apparently they did not work as specified.

COPY - CATS

From the 2014 report, many enthusiasts have attempted to replicate Rossi's generator. Prof Alexander Parkhomov of Lomonosov Moscow State University claims to have accomplished that. The Russian scientist has an established status and unlike Rossi, the good doctor has published a paper, listing equipment and materials. The link contains the translated Russian paper and a video of his demonstration. Alexander's cloned E-Cat was able to get an output, 2.74 times the input which although not as tall as Rossi's claims is still a success. His success sparked a change in the status quo of LENR research.

The Martin Fleischmann Memorial Project (MFMP) is a group dedicated to researching cold fusion reactions and have adopted an open source method to go about it. All their data, procedures and results are available on their website for others to use. Their goal is to facilitate the wide-spread replication and validation of LENR experiments. In February, last year, they were able to get a high burst of gamma rays in their generator, confirming the occurrence of low energy nuclear reactions. Bob Gree-nyer from MFMP stated, "It would appear from our initial findings that Rossi has exactly what he claims." Research on LENR: <http://dgit.in/BENrgyCrp>

STILL A HOAX?

"LENR is real experimentally, and not understood theoretically. There are results that you just can't explain away. Whether it's cold fusion, low-energy nuclear reactions, or something else, we still don't know. But there's no doubt that you can trigger nuclear reactions using chemical energy," told Dr. David Nagel, a professor from George Washington University to C&EN. The current status of LENR research cannot have been put more beautifully than Dr. David Nagel. There have been countless experiments on LENR that have given results that are undeniable. Established institutions are now giving LENR a second chance. It's only a matter of time before a solid scientific theory is established. Coming back to the E-Cat, Rossi might have a bit of a spotty history, but the success of independent research teams proves he's not just all talk. Rossi has made extraordinary claims. But extraordinary claims require extraordinary proof which is why we remain sceptical about the E-cat. But LENR as a field is outgrowing its toxicity. 🚩

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